

Assignment

Question-1

a) Perspective projection is considered more realistic in 3D graphics because it mimics human vision, objects appear smaller as they get farther, and parallel lines converge, giving a sense of depth. This makes 3D graphics in games and movies look natural. Parallel projection, however, keeps object sizes and parallel lines unchanged, allowing accurate measurements.

b) Here, $P(10, 20, -40)$; $\beta = 30^\circ$; $\lambda = 1$

for xy plane —

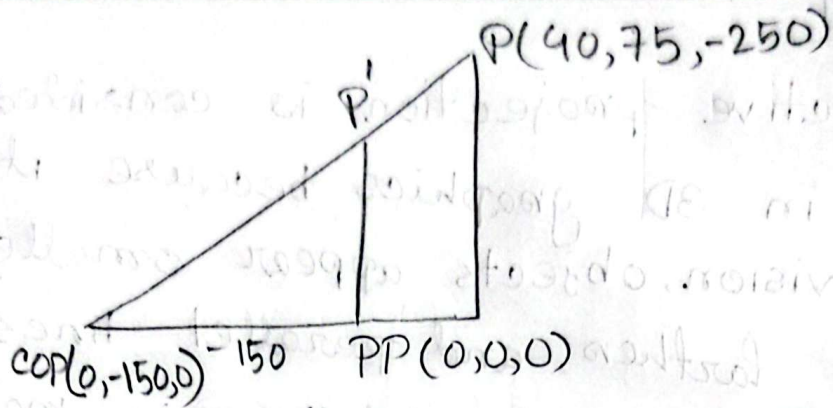
$$P' = \begin{bmatrix} 1 & 0 & \lambda \cos \beta & 0 \\ 0 & 1 & \lambda \sin \beta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \sqrt{3}/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -24.64 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

(Ans)

Q



$$P' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1/150 & 0 & 1 \end{bmatrix} \begin{bmatrix} 40 \\ 75 \\ -250 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3/2} \begin{bmatrix} 40 \\ 0 \\ -250 \\ 3/2 \end{bmatrix}$$

$$= \begin{bmatrix} 26.67 \\ 0 \\ -166.67 \\ 1 \end{bmatrix}$$

$$\therefore P' = (26.67, 0, -166.67)$$

(Ans)

(wri)

Question-2

a) This is cavalier projection. Because it has —

- ① The reading is drawn without any scale reduction
- ② One axis is drawn at oblique angle.

* If the length axis were reduced to half its actual length, then it would become a cabinet projection matrix.

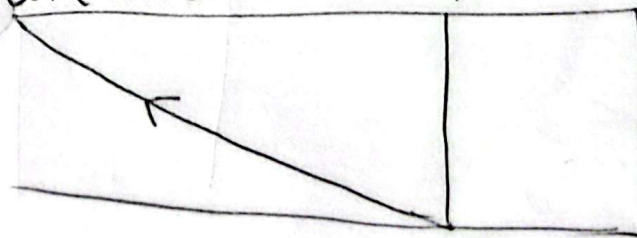
b) Given $A(200, 150, 360)$ &
 $B(600, 130, 360)$

$$P = A + 0.4(B - A)$$

$$= (200, 150, 360) + 0.4[(600 - 200), (130 - 150), (360 - 360)]$$

$$= (360, 142, 360)$$

$$COP(180, 300, 40) \quad P'$$



$$PP(0, 0, 120)$$

$$PP = (0, 0, 120)$$

$$d = \frac{1}{3} (360 - 120) = 80 \quad | \quad z_P = 120,$$

$$\therefore COP = (180, 300, 120 - 80) \\ = (180, 300, 40)$$

$$q(q_x, q_y, q_z) = COP - PP \\ = (180, 300, -80)$$

$$P' = \begin{bmatrix} 1 & 0 & \frac{180}{80} & 120\left(\frac{180}{-80}\right) \\ 0 & 1 & \frac{300}{80} & 120\left(\frac{300}{-80}\right) \\ 0 & 0 & \frac{-120}{80} & 120 + \left(\frac{120}{-80}\right) \\ 0 & 0 & -\frac{1}{80} & \left(1 + \frac{120}{-80}\right) \end{bmatrix} \begin{bmatrix} 360 \\ 142 \\ 360 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 900 \\ 1842 \\ 480 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 225 \\ 260.5 \\ 120 \\ 1 \end{bmatrix}$$

$$\therefore P' = (225, 260.5, 120)$$

(Ans)